SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct

I **P D inesh Reddy / 192525399** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-2

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Title

Create a Simple Cloud Software Application for Hospital Information System Using Google AppSheet (SaaS)

Aim

To develop a cloud-based Hospital Information System using Google AppSheet to demonstrate the Software as a Service (SaaS) model.

Objective

- To understand the SaaS model in cloud computing.
- To create a Hospital Information System using Google AppSheet.
- To store patient information securely in the cloud.
- To manage hospital records efficiently using Google Sheets.

Introduction

A Hospital Information System (HIS) is a software application used to managepatient information, doctor details, and hospital records. By using cloud computing, hospital data can be stored online and accessed securely from anywhere. Google AppSheet provides a no-code platform for developing cloud-based applications, making it suitable for creating a simple Hospital Information System.

Cloud Service Provider

Google AppSheet

Software Requirements

- Google Account
- Google AppSheet
- Google Sheets
- Google Chrome Browser
- Internet Connection

Hardware Requirements

- Laptop/Desktop
- Minimum 4 GB RAM

- Stable Internet Connection

Required Fields

The Hospital Information System should include the following fields:

- Patient Name
- Age
- Address
- Phone Number
- Email ID
- Attender Name
- Attender Phone Number
- Doctor Name
- Disease/Diagnosis

Sample Patient Records

Patient Name	Age	Doctor	Disease	Phone
Ravi Kumar	45	Dr. Kumar	Fever	9876543210
Priya Sharma	30	Dr. Meena	Diabetes	9123456789
Arjun Reddy	52	Dr. Suresh	Hypertension	9988776655
Sneha	26	Dr. Anitha	Migraine	9090909090

Expected Output

The application allows hospital staff to enter patient details through a cloud-based form. The submitted information is automatically stored in Google Sheets and can be viewed, updated, and managed from any authorized device connected to the Internet.

Advantages

- Easy patient record management
- Cloud-based data storage
- Accessible from anywhere
- Secure and reliable
- No software installation required
- Easy to update patient information

Result

A cloud-based Hospital Information System was successfully created using Google AppSheet. Patient information was entered through the application and stored securely in Google Sheets, demonstrating the SaaS model.

Conclusion

The Hospital Information System developed using Google AppSheet simplifies patient data management through cloud technology. The application provides secure storage, easy accessibility, and efficient record management, making it a practical example of Software as a Service (SaaS).

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

.

Title

Create a Simple Cloud Software Application for an Online Supermarket Using Google AppSheet (SaaS)

Aim

To develop a cloud-based Online Supermarket application using Google AppSheet that demonstrates the Software as a Service (SaaS) model for managing customer orders and product details.

Objective

- To understand the SaaS model in cloud computing.
- To create an Online Supermarket application using Google AppSheet.
- To collect customer details and product information.
- To store order details securely in Google Sheets.
- To enable online order management through a cloud-based application.

Introduction

An Online Supermarket System is a cloud-based application that allows customers to purchase grocery items through an online platform. Customers can enter their personal information, select product categories, choose items, and specify the required quantity. Google AppSheet is a no-code platform that simplifies the development of such applications by connecting directly to Google Sheets, where all order data is stored and managed.

Cloud Service Provider
Google AppSheet

Software Requirements

- Google Account
- Google AppSheet
- Google Sheets
- Google Chrome Browser
- Internet Connection

Hardware Requirements

- Laptop/Desktop
- Minimum 4 GB RAM
- Stable Internet Connection

Required Fields

The Online Supermarket application should contain the following fields:

- Customer Name
- Address
- Phone Number
- Email ID
- Item Category
- Item Name
- Quantity

Item Classification

Category	**Items**
Fruits	Apple, Banana, Orange
Vegetables	Tomato, Potato, Onion
Dairy	Milk, Butter, Cheese
Bakery	Bread, Cake, Bun
Snacks	Chips, Biscuits, Chocolate

Category	Items
Beverages	Tea, Coffee, Juice
Household	Soap, Shampoo, Detergent

Procedure

Open Google Sheets and create a new spreadsheet.
Add columns for Customer Name, Address, Phone Number, Email, Item Category, Item Name, and Quantity.
Open Google AppSheet and create a new application.
Connect the Google Sheet as the data source.
Design a customer order form with the required fields.
Add a dropdown list for Item Category.
Save and deploy the application.
Enter sample customer orders and verify that the data is stored in Google Sheets.

Sample Customer Records

Customer Name	Category	Item	Quantity	Phone
Dinesh Reddy	Fruits	Apple	3	9876543210
Rahul Kumar	Dairy	Milk	2	9988776655
Priya Sharma	Snacks	Chips	5	9123456789
Arjun	Bakery	Bread	1	9090909090

Expected Output

The customer enters personal details, selects an item category, chooses a product, enters the required quantity, and submits the order. The order information is automatically stored in Google Sheets and can be accessed through the AppSheet application.

Advantages

Easy online ordering process
Cloud-based data storage
Accessible from anywhere
Automatic data synchronization

No programming knowledge required
Fast and secure order management

Result

The Online Supermarket application was successfully created using Google AppSheet. Customer order details were stored in Google Sheets, demonstrating the Software as a Service (SaaS) model through cloud computing.

Conclusion

The Online Supermarket application provides an efficient method for managing customer orders using cloud technology. By integrating Google AppSheet with Google Sheets, the application enables secure data storage, real-time updates, and easy accessibility, demonstrating the practical implementation of the SaaS model.

Experiment 3:

Demonstrate virtualization by Installing Type-Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Title

Demonstrate Virtualization by Installing a Type-2 Hypervisor and Creating a Virtual Machine with 3 CPU, 10 GB RAM, and 10 GB Storage

Aim

To demonstrate virtualization by installing a Type-2 Hypervisor and creating a Virtual Machine (VM) with 3 CPU cores, 10 GB RAM, and 10 GB virtual hard disk.

Objective

- To understand the concept of virtualization.
- To install a Type-2 Hypervisor on the host operating system.
- To create and configure a Virtual Machine.
- To allocate CPU, memory, and storage resources to the VM.

- To understand the benefits of virtualization in cloud computing.

Introduction

Virtualization is the process of creating virtual versions of computing resources such as servers, storage, and operating systems. A **Type-2 Hypervisor** runs on top of an existing host operating system and allows multiple virtual machines to run on a single computer. Popular Type-2 hypervisors include Oracle VirtualBox and VMware Workstation. Virtualization is widely used in cloud computing to improve resource utilization, reduce hardware costs, and simplify system management.

Software Requirements

- Oracle VirtualBox or VMware Workstation
- Windows or Linux Host Operating System
- Windows/Linux ISO Image
- Internet Connection (optional)

Hardware Requirements

- Laptop/Desktop Computer
- Minimum Intel Core i5 or equivalent processor
- 16 GB RAM (recommended)
- 30 GB Free Disk Space

Procedure

1. Download and install Oracle VirtualBox or VMware Workstation.
2. Launch the virtualization software.
3. Click **New** to create a Virtual Machine.
4. Enter the virtual machine name and select the operating system type.
5. Allocate **3 CPU cores** to the virtual machine.
6. Allocate **10 GB RAM**.

7. Create a **10 GB Virtual Hard Disk**.
8. Attach the operating system ISO file.
9. Start the virtual machine.
10. Install the guest operating system.
11. Verify that the virtual machine boots successfully.

Virtual Machine Configuration

Parameter	**Configuration**
Hypervisor	Oracle VirtualBox
Guest Operating System	Ubuntu 22.04 LTS
CPU	3 Cores
RAM	10 GB
Storage	10 GB Virtual Disk
Network	NAT

Expected Output

A virtual machine is successfully created and configured with **3 CPU cores**, **10 GB RAM**, and a **10 GB virtual hard disk**. The guest operating system installs and runs independently on the host machine through the Type-2 Hypervisor.

Advantages

- Efficient utilization of hardware resources.
- Multiple operating systems can run on one computer.
- Easy testing and software development.
- Reduced hardware costs.
- Improved system isolation and security.
- Simplified backup and recovery.

Applications

- Cloud Computing
- Software Development
- Testing and Debugging
- Server Consolidation
- Educational Laboratories
- Disaster Recovery

Result

A Type-2 Hypervisor was successfully installed, and a Virtual Machine was created with **3 CPU cores, 10 GB RAM, and a 10 GB virtual hard disk**. The guest operating system was installed successfully, demonstrating the concept of virtualization.

Conclusion

The experiment successfully demonstrated virtualization using a Type-2 Hypervisor. The virtual machine was configured with the specified CPU, memory, and storage resources and operated independently of the host system. This experiment illustrates how virtualization supports efficient resource utilization and forms the foundation of modern cloud computing environments.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

To demonstrate virtualization by creating a snapshot of a Virtual Machine (VM) using a Type-2 Hypervisor and restoring the VM to its previous state.

Objective

- To understand the concept of VM snapshots.
- To create a snapshot of a virtual machine.
- To modify the virtual machine after taking a snapshot.
- To restore the VM to its previous state using the snapshot.
- To understand the importance of snapshots in virtualization.

Introduction

A Virtual Machine (VM) snapshot is a saved state of a virtual machine at a specific point in time. It captures the VM's operating system, applications, settings, and data. If any problem occurs after making changes, the snapshot can be restored to return the VM to its previous working condition. Snapshots are widely used for software testing, system updates, backup, and disaster recovery.

Software Requirements

- Oracle VirtualBox or VMware Workstation
- Windows or Linux Host Operating System
- Existing Virtual Machine
- Internet Connection (optional)

Hardware Requirements

- Laptop/Desktop Computer
- Minimum 8 GB RAM
- Intel Core i5 or equivalent processor
- Sufficient Free Disk Space

Procedure

1. Open Oracle VirtualBox or VMware Workstation.

2. Select the existing Virtual Machine.
3. Start the virtual machine and ensure it is working properly.
4. Shut down the VM if required by the hypervisor.
5. Select the **Snapshots** option.
6. Click **Take Snapshot** and provide a name such as **Initial State**.
7. Start the VM and make some changes, such as creating a folder or installing software.
8. Shut down the virtual machine.
9. Open the **Snapshots** menu.
10. Select the previously created snapshot.
11. Click **Restore Snapshot**.
12. Start the VM again and verify that the previous state has been restored successfully.

Snapshot Details

Parameter	**Description**
Hypervisor	Oracle VirtualBox
Snapshot Name	Initial State
Guest Operating System	Ubuntu 22.04 LTS
Purpose	Backup and Recovery
Status	Successfully Restored

Expected Output

A snapshot of the virtual machine is created successfully. After making changes to the VM, restoring the snapshot returns the virtual machine to its original state, proving that the snapshot feature works correctly.

Advantages

- Quick backup of the virtual machine.
- Easy restoration after system failure.
- Safe environment for software testing.
- Saves time during recovery.
- Protects important system configurations.
- Supports disaster recovery.

Applications

- Software Testing
- System Maintenance
- Backup and Recovery
- Educational Laboratories
- Cloud Infrastructure
- Server Management

Result

A snapshot of the virtual machine was successfully created using the Type-2 Hypervisor. After making changes to the virtual machine, the snapshot was restored successfully, and the VM returned to its previous working state.

Conclusion

The experiment successfully demonstrated the snapshot feature available in a Type-2 Hypervisor. Creating and restoring a snapshot allowed the virtual machine to recover its earlier state without reinstalling the operating system or applications. This feature is an essential component of virtualization and is widely used in cloud computing environments for backup, testing, and disaster recovery.